

BUSINESS REQUIREMENTS DOCUMENT (BRD)

LifeBook AI — Multimodal Digital Diary & Personal Life Assistant

Formal Business Specification for Cognitive Memory Capture, Personal Life Graph Synthesis, and Proactive Assistant Intelligence.

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Target Stakeholders	Product Engineering, AI Architecture, UX/UI Design, Executive Review

EXECUTIVE MANDATE

In an era of acute digital fragmentation, human knowledge workers and creatives experience significant memory decay and cognitive overload. Contemporary journaling tools demand high cognitive tax and manual categorization, resulting in abandoned logs and lost life context. LifeBook AI bridges this gap by combining zero-friction multimodal capture (voice, photos, text) with an autonomous 11-stage background pipeline that constructs an interconnected Personal Life Graph. It delivers actionable commitment tracking, proactive routine deviation detection, and multi-tenant AI conversational assistance while strictly preserving privacy.

1. Business Background & Problem Statement

1.1 Market & Cognitive Problem Analysis

Traditional journaling and personal knowledge management (PKM) solutions suffer from four fundamental failure modes:

- **High Input Friction:** Manual typing after an exhausting day creates unsustainable friction. Users fail to maintain consistency beyond 14 consecutive days.
- **Siloed Multimodality:** Photographs remain buried in cloud camera rolls; fleeting voice ideas remain in voice memos without synthesis; physical handwritten thoughts are rarely transcribed.
- **Dead Archives (Digital Amnesia):** Static chronological diary notes are rarely revisited or cross-referenced. Meaningful connections between people, recurring places, and ongoing projects are permanently lost.
- **Passive Logging without Proactive Value:** Existing diaries record what happened in the past but provide zero forward-looking utility—such as detecting that an implicit promise made to a colleague is past due.

1.2 The LifeBook AI Vision

LifeBook AI transforms the age-old leather journal into a living, intelligent personal assistant. By marrying a tactile, warm aesthetic (leather textures, creamy parchment, gold foil accents) with state-of-the-art AI orchestration, it preserves life memories effortlessly while unlocking proactive intelligence for high-performing individuals.

2. Target User Personas & Use Cases

Persona	Profile & Context	Primary Pain Points	LifeBook AI Solution Value
Joe_Dev / Jothiram CS-Math Student & Creative Artist	Multidisciplinary builder balancing algorithms, hackathons, and portrait art projects. Captures spontaneous voice thoughts and reference photos.	Scattered project notes; misses implicit verbal commitments made during rapid hackathons; forgets artwork progress milestones.	Instant voice dictation normalization; visual Polaroid gallery for portraits; automated commitment extraction with deadline reminders.
Dr. Elena Vance Senior Research Director	Manages multiple research grants, laboratory staff, and international conference keynotes. High cognitive load, strict confidential data needs.	Information overload across meetings; lack of time to write prose journals; strict privacy needs preclude public cloud training.	Speech-to-text pipeline transcribes walk-and-talk voice debriefs; Life Graph connects researchers to lab equipment and publication topics; local-first privacy.
Marcus Chen Travel Writer & Founder	Mobile-first professional moving across cities, capturing architectural photos, café inspirations, and casual meetings.	Photos disconnected from written reflections; forgets names of acquaintances met at regional hubs.	Scrapbook photos view automatically categorized; People & Places directory in radial graph links encounters with geotagged locations.

3. Business Functional Requirements (BFR)

Requirements are classified as **P0 (Core MVP - Mandatory)**, **P1 (High Priority)**, and **P2 (Strategic Growth)**.

ID	Requirement Title	Pr i	Functional Business Scope & Acceptance Criteria
BFR-01	Multimodal Input Ingestion	P0	The system shall accept simultaneous or independent inputs of audio voice recordings, high-resolution photographs, and raw text notes without forcing pre-formatting. Durability guarantee: raw inputs must persist before AI processing commences.
BFR-02	Durability-First Processing Pipeline	P0	Every entry must execute an atomic 11-stage state machine from CREATED to USER_REVIEW. Background jobs must run decoupled from web requests, and failed steps must be re-triable on demand without data loss.
BFR-03	AI Narrative Diary Synthesis	P0	The AI engine shall transform fragmented voice transcripts and bullet notes into a warm, reflective first-person diary entry. The user can review, edit, and confirm the synthesized memory before permanent archival.
BFR-04	Personal Life Graph & Entity Extraction	P0	The system must extract structured entities (People, Places, Activities, Projects) and map bidirectional relationships. Users shall interactively explore relationships via a dynamic radial graph and dedicated People/Places directories.
BFR-05	Commitment & Promise Extraction	P0	The system must detect explicit and implicit promises (e.g., 'Need to send portrait preview to Joe on Friday'), extract due dates, identify the target counterparty, and display them in an actionable commitment tracker.
BFR-06	Habit & Routine Deviation Monitoring	P1	The engine shall identify recurring behavioral routines (e.g., daily math problem solving, gym sessions) and trigger gentle alerts when habitual cadence drops below historical baseline.
BFR-07	Multi-Tenant Contextual Assistant	P0	Users must have an AI conversation assistant powered by OpenRouter LLM that answers questions strictly from the authenticated user's private diary memories, profile, and knowledge graph without cross-tenant bleed.
BFR-08	Polaroid Scrapbook Media Gallery	P1	Photos attached to diary entries must be cataloged in a visual scrapbook modal with filter tabs (All, Memories, Projects, Travel) and high-resolution modal inspection.
BFR-09	Data Privacy & Local Sovereignty	P0	User diary data must never be transmitted to public LLMs for foundation training. Offline fallback AI providers must be available when cloud API connectivity is interrupted.

4. Non-Functional Requirements (NFR)

Category	Requirement & Metric Target	Validation Methodology
Latency & Performance	Ingestion acknowledgement: < 300 ms. Full 11-stage processing cycle: < 6.0 seconds for voice + photo + text.	Automated load testing with simulated 60s audio uploads and 5MB image payloads.
Reliability & Durability	Zero data loss on server crash. If background worker terminates mid-pipeline, state must resume or offer 1-click retry.	Kill process test during STT stage; verify retry restores pipeline to USER_REVIEW.
Security & Isolation	Argon2 / bcrypt password hashing; stateless JWT session tokens; multi-tenant database foreign key isolation; SQL injection prevention.	Security audit test suite ensuring User B cannot read User A entities or queries.
User Experience & Aesthetics	Warm leather journal aesthetic, typography pairing (Playfair Display / Serif + Clean Sans), responsive mobile/tablet layout.	Lighthouse Accessibility score >= 92; visual regression testing on desktop and mobile viewports.

5. Business Value, Key Performance Indicators (KPIs) & Roadmap

5.1 Success Metrics & Target KPIs

Metric	Baseline / Industry Avg	LifeBook AI Target	Business Impact
Daily Journaling Retention	12% at 30 days (Manual apps)	>= 58% at 30 days	Frictionless voice input removes the primary cause of user abandonment.
Commitment Fulfillment	45% verbal commitments forgotten	>= 92% tracked & alerted	Increases user productivity, professional reliability, and trust in the assistant.
Time-to-Log Memory	4.5 minutes (Typing on mobile)	< 35 seconds (Speak & snap)	Enables on-the-go capture during transit, walks, and immediate project debriefs.
Assistant Query Accuracy	Generic LLM hallucination rate ~20%	> 98% factual precision	Grounded retrieval directly from user's confirmed entries and Life Graph.

5.2 Phased Strategic Roadmap

Phase	Timeline	Key Deliverables & Milestones
Phase 1: Foundation	Completed (Q1)	Multimodal ingestion, SQLite persistence, Durability pipeline, React leather journal UI, basic entity tagging.
Phase 2: Intelligence	Current Baseline	11-stage state machine, OpenRouter gpt-4o-mini integration, Life Graph radial visualizer, Polaroid photo gallery, commitment tracking.
Phase 3: Proactive AI	Target: Q3 2026	Autonomous morning audio briefings, predictive commitment scheduling, biometric stress correlation, local Llama-3 ONNX runtime.
Phase 4: Ecosystem	Target: Q4 2026	Encrypted multi-device sync, smart stylus handwriting OCR support, export to physical leather bound book printing.

SIGN-OFF & GOVERNANCE

The functional and business requirements specified in this document have been reviewed and approved by the Product, Architecture, and Engineering stakeholders. Any material scope changes must follow formal change control procedures.